

Retail Business Performance & Profit Optimization Analysis

Tools Used: Excel, SQL, Power BI

Project Type: Business/Data Analytics Portfolio Project

1. Executive Summary

This project focuses on analyzing retail sales and profitability data to identify revenue drivers, loss-making areas, customer behavior trends, and opportunities for business optimization. The project was developed using Excel for initial analysis and cleaning, SQL for business querying and KPI extraction, and Power BI for interactive dashboard development and business storytelling.

2. Business Problem Statement

Retail businesses often generate high sales without maintaining healthy profitability. The company needed a detailed analysis to identify which categories, products, customer segments, and regions were contributing positively or negatively to overall business performance.

3. Project Objectives

- Analyze overall sales and profit performance.
- Identify high-performing and loss-making products.
- Understand regional and customer segment contribution.
- Measure discount impact on profitability.
- Create interactive dashboards for decision-making.
- Provide recommendations for profit optimization.

4. Dataset Overview

The dataset contains retail sales transaction records including order details, product information, customer segments, shipping modes, regions, sales, discounts, quantities, and profits.

Key Columns	Description
Sales	Revenue generated from orders
Profit	Net profit earned
Discount	Discount applied on orders
Category/Sub-category	Product classification

Region/State	Geographical information
Segment	Customer segment
Order Date	Order timeline

Dataset Size: 9,994 rows and 21 columns

Time Period: 2014–2017

Dataset Type: Retail Sales Transaction Data

5. Excel Analysis & Preparation

- Reviewed and structured raw retail dataset.
- Performed basic data cleaning and formatting.
- Verified missing values and duplicate records.
- Created summary analysis using formulas and pivot tables.
- Prepared dataset structure for SQL and Power BI.

6. SQL Analysis

SQL was used for business querying, KPI analysis, customer segmentation analysis, regional performance evaluation, and profitability assessment.

- Data overview and quality checks
- Total sales, total profit, and average discount calculations
- Category-wise and sub-category-wise performance analysis
- Top 10 products by sales
- Customer segment performance analysis
- Regional sales and profitability analysis
- Profit margin analysis
- Identification of high-sales but low-profit products
- Discount impact evaluation

7. Power BI Development

7.1 Data Modeling

- Created a star schema model.
- Built Fact_Sales and multiple dimension tables.
- Created a dedicated Date table using DAX.
- Established optimized one-to-many relationships.
- Organized measures inside a dedicated Measures table.

7.2 DAX Measures Created

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Profit Margin %
- Average Discount
- Sales Per Order
- Profit Per Order
- Previous Year Sales
- YoY Sales Growth %
- Running Total Sales
- Loss Transactions
- Loss %

7.3 Dashboard Pages

- Executive Overview
- Sales Analysis
- Profitability Analysis
- Customer & Regional Insights
- Profit Optimization

8. Key Business Insights

- Some product categories generated strong sales but comparatively lower profit margins.
- High discount levels negatively impacted profitability in several product groups.
- Certain products consistently generated losses despite good sales volume.
- Regional performance varied significantly across the business.
- Customer segments contributed differently to overall profitability.
- A few sub-categories required pricing and discount optimization.

9. Business Recommendations

- Reduce excessive discounts on low-margin products.
- Focus more marketing efforts on profitable product categories.
- Review pricing strategies for loss-making products.
- Improve inventory planning for high-performing categories.
- Use region-specific sales strategies for weak-performing areas.
- Monitor profit margin trends continuously using BI dashboards.

10. Conclusion

This project successfully transformed raw retail sales data into a business intelligence solution capable of supporting strategic decision-making. Using Excel, SQL, and Power BI, the project demonstrated data cleaning, querying, modeling, KPI development, dashboard design, and insight generation skills. The final dashboard enables business users to monitor sales, profitability, customer trends, and optimization opportunities effectively.

11. Skills Demonstrated

- Data Cleaning
- Business Analysis
- SQL Querying
- Power BI Dashboarding
- DAX Calculations
- Data Modeling
- KPI Development
- Data Visualization
- Business Storytelling

12. Resume Project Description

Developed a multi-page Retail Business Performance & Profit Optimization dashboard using Excel, SQL, and Power BI. Performed data cleaning, business KPI analysis, star schema modeling, DAX measure creation, and interactive dashboard development to analyze sales, profitability, customer segments, regional trends, and optimization opportunities.